

Manmohan Kinthali

Embedded Software Engineer

Eight+ years of experience in the embedded industry with ISO26262 L1 Certification. Involved in the design and development of various systems and proficient in various platforms, for White goods, Radar Engineering and Power Electronics domains.



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Braunschweig, Germany

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WORK EXPERIENCE

Software Engineer

Smart Microwave Sensors GmbH

10/2022 - Present

Braunschweig, Germany

Achievements/Tasks

- Design and verification of Software Architecture.
- Preparation of SW-Cs and integration into BSW.
- Implementation of Radar Frontend algorithm for mmWave
- System design considering Memory Management and Runtime optimization for SPC & Tracking Modules.
- Planning & Coordination of Product developments.
- Safety Analysis of SW components in context of ISO-26262.
- Responsible for Baseline SW, and FuSa implementations.
- Responsible for Platform Problem Management.

Specialist

ZF Technical Center India Pvt. Ltd.

08/2019 - 09/2022

Hyderabad, India

Achievements/Tasks

- Develop, analysis & validation of Architectural Design.
- Design & Development of various systems and algorithms in Power Electronics for Hybrid and Battery EVs.
- Impact analysis on systems, incl. FMEA and FTA.
- Specification and verification of HSI, SW and FuSa Req's.
- Developed RAD Master which enables the discharge path of DC link capacitors in OFF state. Maintaining the droop until voltage reaches to safe level being a big challenge.
- Active participation in reviews, ASPICE & FuSa assessments.

Embedded Programmer

Invent Solutions Pvt. Ltd.

06/2017 - 08/2019

Chennai, India

Achievements/Tasks

- Implementation of sub systems in EVs and white goods.
- Research and Developing strategies for EV Motor Control.
- Gone through complete Project life cycle for EV-1kW.
- Tuned BLDC using FOC and speed profiles for Urea pump.
- Delivered a desert cooler to Bajaj, whole product life cycle.
- Working on PoCs and Prototypes for PMSM, BLDC, sensor, sensor less, open and closed loop motor controlling.
- Implementation of LLD & Configuration of Tuya IoT module.

ACHIEVEMENTS

Secured ISO 26262 Level-1 (05/2023 - Present)

Got L1 certification from TUV SUD.

Got R&R and SPOT Awards (01/2021 - 06/2021)

Got appreciation from the BU Manager for the work products delivered without any defects and also appreciation from the Quality team.

Proposed paper is elected as Univ #1 paper (05/2015 - 06/2015)

The paper describes Optic Sensors which can be used to detect the health status of bridges, and many other structures.

SKILLS

ISO26262

Functional Safety

Git

Enterprise Architect

ASPICE

DOORS

Clear Case

Embedded C

CAN

SPI, UART, I2C

Ethernet

Embedded Design & Development

SW ARCH & Integration

Trace32

MCAL

LLD

SVN

TriCore

Power Electronics

AUTOSAR

Jenkins

C

Medini analyze

Radar Processing Chain

SAFEAgile

Debugging

Power PC

Blackhawk

RCA

Python

PROJECTS

Race Runner Ultra 2 - NXP S32R294 (10/2022 - Present)

- Responsible for Baseline SW, also in context of FuSa ISO26262
- Safety analysis and implementations for the platform
- Creation of SwAD, Implementation of SW-Cs, and validation

Multi and Plug-in HEV (08/2019 - 09/2022)

- Implemented multiple features and rational checks for power electronic components incl. Safety Analysis
- Responsible for Temperatures signal conditioning for project
- Configurations of External ADC chip for HV Measurement

Battery EV (FEV) (08/2019 - 09/2022)

- Design and Implementation of power electronic systems
- Responsible for Problem analysis and resolution; and Functional Safety owner for Emergency Stop, Temperatures and External ADC
- Active member for ASPICE assessment for the project

BSG Control Unit (03/2018 - 10/2018)

- Implemented torque boost control mechanism
- Designed open and closed loop FOC algorithm
- CAN Master to command the BAS machine for cranking, torque boost, alternator, recuperation modes

Electric Vehicle (06/2018 - 08/2019)

- Implemented Trapezoidal and FOC bi-dirac algorithms using single-shunt current reconstruction
- Design of current injection using throttle positioning
- Integration of BCM, BMS, PE to the APPL

High Voltage BLDC Fan (11/2017 - 03/2018)

- Implementation of device drivers and rotor positioning
- Code optimization to bring down to 8k flash and 1k RAM

EDUCATION

Bachelor of Technology (Elec & Comm)

GMR Institute of Technology, JNTUK, India

09/2013 - 06/2017

LANGUAGES

English

Full Professional Proficiency

German

Limited Working Proficiency